



A Classification Analysis of Wine Quality

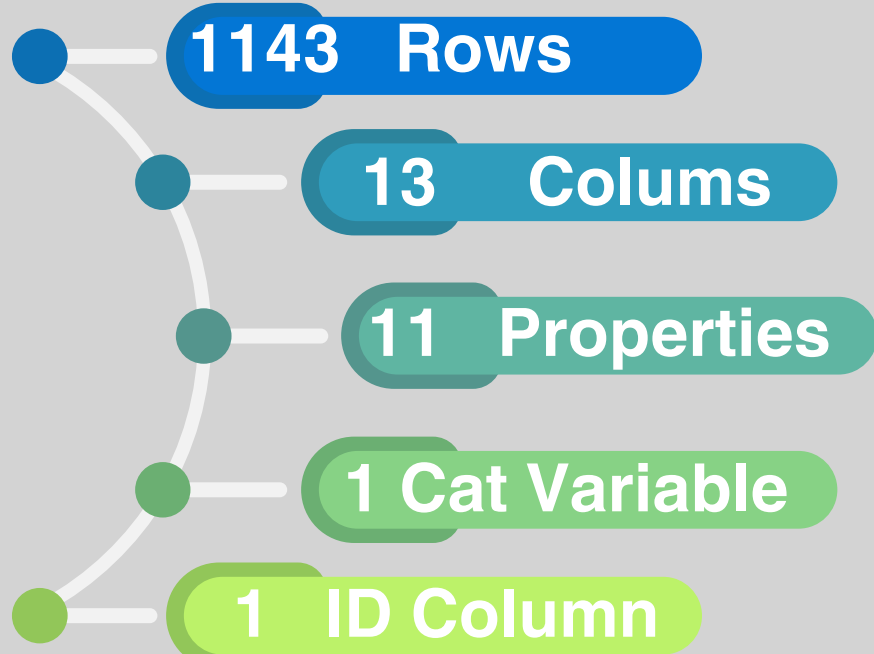
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Using physiochemical properties in wine quality prediction

A. Data Overview

The dataset pertains to red variants of Portuguese 'Vinho Verde' wine.



B. Business Use

- ✓ Automated Wine Quality Prediction.
- ✓ Optimizing Production Processes.
- ✓ Improving Pricing Decisions
- ✓ Product Segmentation

C. Methodology

EDA Analysis

Univariate + Bivariate Analysis

Multidimensionality Reduction

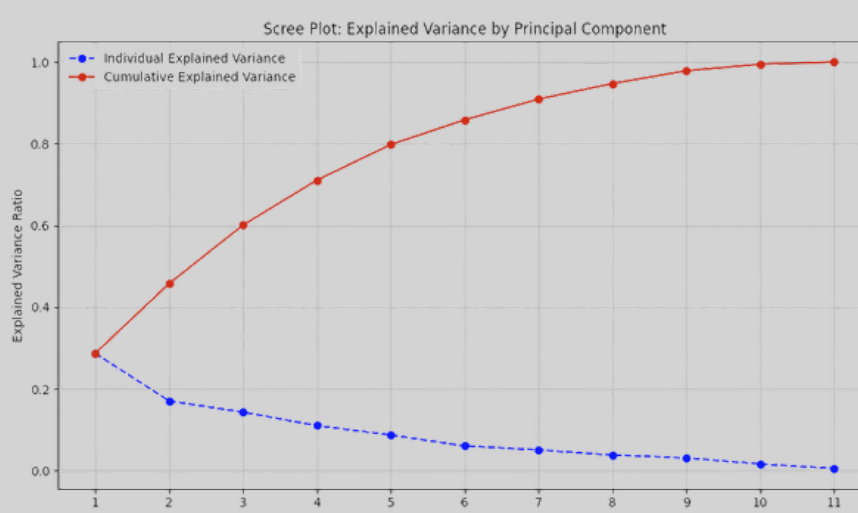
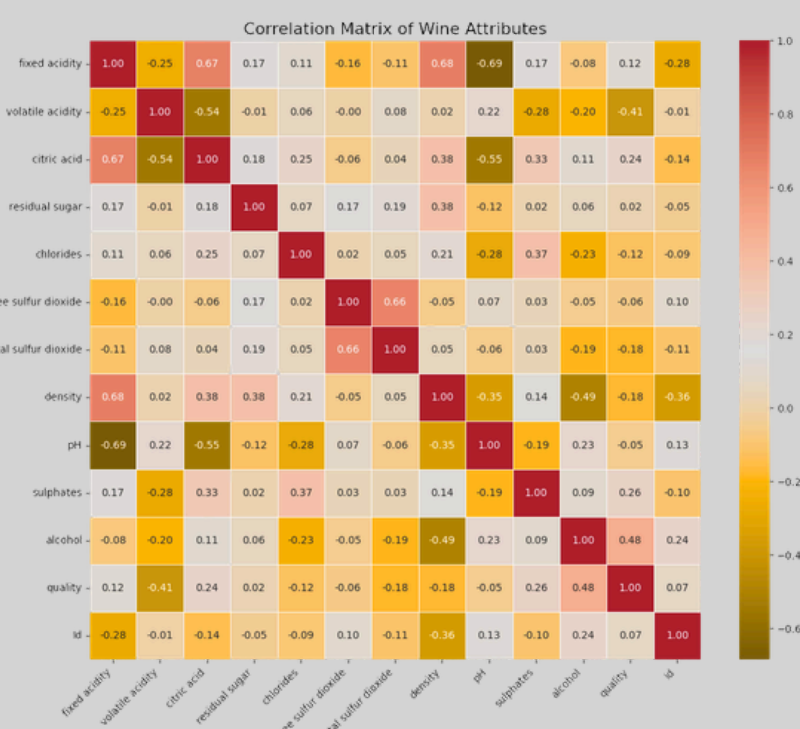
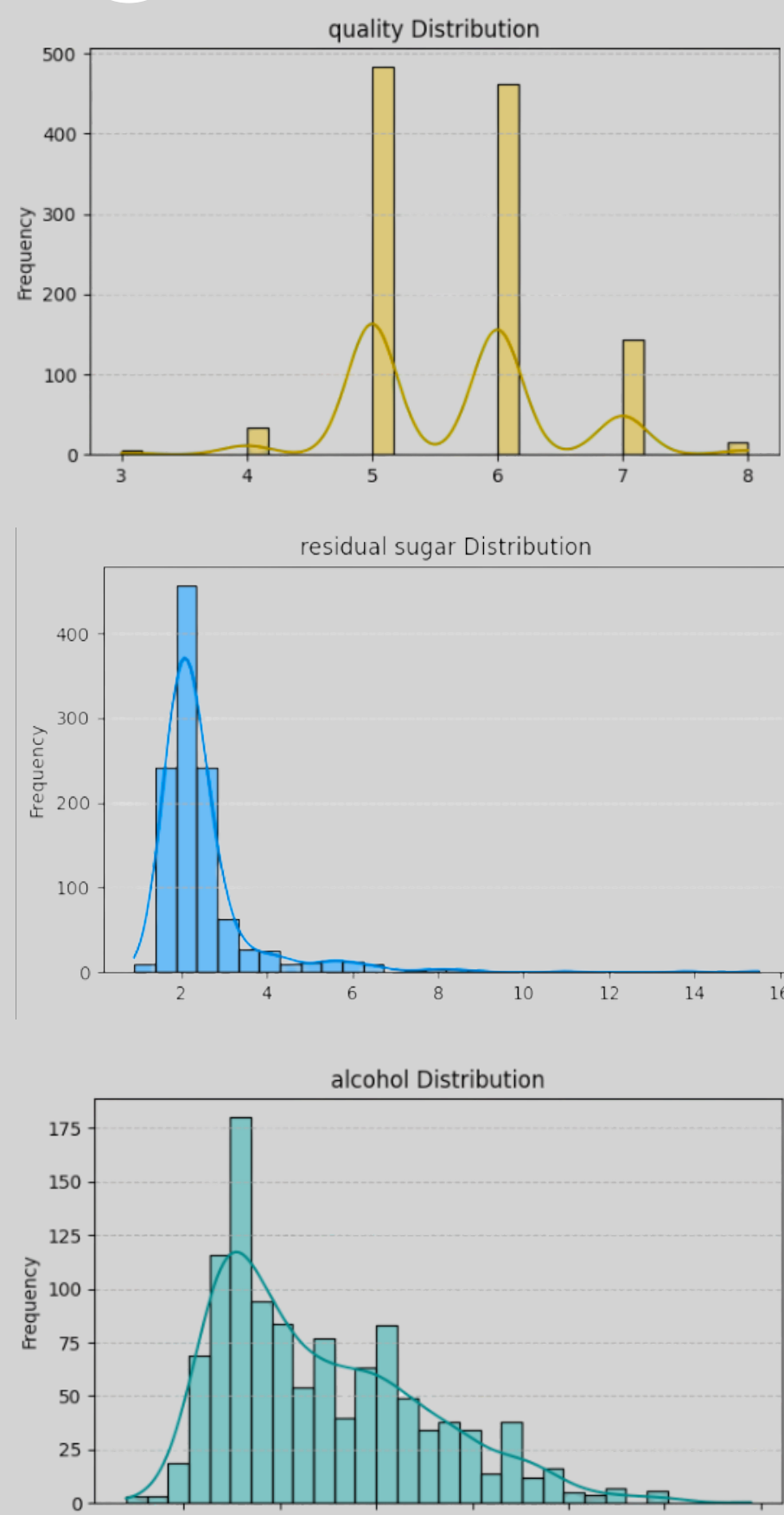
PCA Analysis

PCA performed successfully.
Shape of principal components: (1143, 11)

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	PC11
0	-1.561545	0.444892	-1.767209	0.132635	0.130138	0.906815	0.122336	0.330086	-0.011548	-0.249611	0.046465
1	-0.760892	1.915058	-0.864158	0.536890	0.090935	-0.860406	0.998623	-0.746939	0.547644	0.005196	-0.134810
2	-0.708085	0.914423	-1.142363	0.435075	0.024940	-0.356702	0.536162	-0.575652	0.084118	-0.203715	-0.122553

Explained Variance Ratio	Cumulative Explained Variance:
PC1: 0.2869	PC1: 0.2869
PC2: 0.1708	PC2: 0.4577
PC3: 0.1431	PC3: 0.6007
PC4: 0.1101	PC4: 0.7109
PC5: 0.0871	PC5: 0.7980
PC6: 0.0605	PC6: 0.8585
PC7: 0.0507	PC7: 0.9092
PC8: 0.0381	PC8: 0.9473
PC9: 0.0313	PC9: 0.9786
PC10: 0.0161	PC10: 0.9947
PC11: 0.0053	PC11: 1.0000

D. Visualisations



E. Findings

1. Distribution of Quality is imbalanced.
2. Distribution of alcohol is irregular, but no pronounced outliers.
3. No missing values.

F. Correlation Analysis

1. Citric Acid & Fixed Acidity (0.67).
2. Alcohol & Quality (0.48).
3. Free Sulphur Dioxide and Total Sulphur Dioxide (0.67).

G. Interpretation

Principal Component Loadings:

	PC1	PC2	PC3	PC4
fixed acidity	0.485339	-0.102328	-0.122376	-0.236046
volatile acidity	-0.227143	0.288786	-0.443718	0.077288
citric acid	0.460075	-0.146508	0.246843	-0.063818
residual sugar	0.174506	0.252262	0.091359	-0.407263
chlorides	0.224870	0.153133	-0.052562	0.652669
free sulfur dioxide	-0.047852	0.517022	0.428288	-0.050809
total sulfur dioxide	0.015069	0.577393	0.323424	-0.046669
density	0.399684	0.217537	-0.334238	-0.164019
pH	-0.432844	-0.006374	0.059923	-0.006724
sulphates	0.237555	-0.062280	0.302768	0.527547
alcohol	-0.118799	-0.381635	0.470201	-0.179953

#	Column	Non-Null Count	Dtype
0	Acidity & Body	1143 non-null	float64
1	Preservation & Fermentation	1143 non-null	float64
2	Alcohol & Stability	1143 non-null	float64
3	Mineral & Saltiness	1143 non-null	float64

types: float64(4)
memory usage: 35.8 KB

H. Limitations & Recommendation

Limitations :

- ✗ Class imbalance in Quality.
- ✗ Quality is subjective
- ✗ Limited features.

Recommendations :

- ✓ Focus on Key Chemical Drivers.
- ✓ Product Segmentation & Strategy.
- ✓ Enhance Data Strategy.